

React

9. ReactJS-HOL

Objectives

- List the features of ES6
- Explain JavaScript let
- Identify the differences between var and let
- Explain JavaScript const
- Explain ES6 class fundamentals
- Explain ES6 class inheritance
- Define ES6 arrow functions
- Identify set(), map()

In this hands-on lab, you will learn how to:

- Use map() method of ES6
- Apply arrow functions of ES6
- Implement Destructuring features of ES6

Prerequisites

The following is required to complete this hands-on lab:

- js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **60 minutes**.

Create a React Application named "cricketapp" with the following components:

1. ListofPlayers

- Declare an array with 11 players and store details of their names and scores using the map feature of ES6
- Filter the players with scores below 70 using arrow functions of ES6.

1. IndianPlayers

2. Display the Odd Team Player and Even Team players using the Destructuring features of ES6
3. Declare two arrays T20players and RanjiTrophy players and merge the two arrays and display them using the Merge feature of ES6

Display these two components in the same home page using a simple if else in the flag variable.

IndianPlayers.js

```
import React from 'react';

const T20Players = ['First Player', 'Second Player', 'Third Player'];
const RanjiTrophyPlayers = ['Fourth Player', 'Fifth Player', 'Sixth Player'];
export const IndianTeam = [...T20Players, ...RanjiTrophyPlayers];

export const OddPlayers = ({ IndianTeam }) => {
  const [first, , third, , fifth] = IndianTeam;

  return (
    <div>
      <ul>
        <li>First: {first}</li>
        <li>Third: {third}</li>
        <li>Fifth: {fifth}</li>
      </ul>
    </div>
  );
};
```

```
// Component to display even players using array destructuring
export const EvenPlayers = ({ IndianTeam }) => {
  // Destructuring to get the second, fourth, and sixth players (indices 1, 3, 5)
  const [, second, , fourth, , sixth] = IndianTeam;

  return (
    <div>
      <ul>
        <li>Second: {second}</li>
        <li>Fourth: {fourth}</li>
        <li>Sixth: {sixth}</li>
      </ul>
    </div>
  );
};

// Component to display the merged list of Indian players
export const ListofIndianPlayers = ({ IndianPlayers }) => {
  return (
    <ul>
      {IndianPlayers.map((player, index) => (
        <li key={index}>Mr. {player}</li>
      ))}
    </ul>
  );
};
```

ListofPlayers.js

```
import React from 'react';

const ListofPlayers = ({ players }) => {
  return (
    <ul>
      {players.map((player, index) => (
        <li key={index}>
```

```

      Mr. {player.name} <span>{player.score}</span>
    </li>
  )))
</ul>
);
};

export default ListofPlayers;

```

App.js

```

import React from 'react';
import './App.css';
import ListofPlayers from './ListofPlayers';
import { OddPlayers, EvenPlayers, ListofIndianPlayers } from './IndianPlayers';
import { IndianTeam } from './IndianPlayers';

// Data for the 'ListofPlayers' component
const players = [
  { name: 'Jack', score: 50 },
  { name: 'Michael', score: 70 },
  { name: 'John', score: 40 },
  { name: 'Ann', score: 61 },
  { name: 'Elisabeth', score: 61 },
  { name: 'Sachin', score: 95 },
  { name: 'Dhoni', score: 100 },
  { name: 'Virat', score: 84 },
  { name: 'Jadeja', score: 64 },
  { name: 'Raina', score: 75 },
  { name: 'Rohit', score: 80 },
];

function App() {
  // Set the flag to 'true' to show the ListofPlayers component
  // Set the flag to 'false' to show the IndianPlayers component
  const flag = false;

```

```

if (flag === true) {
  const playersBelow70 = players.filter(player => player.score <= 70);
  return (
    <div className="App">
      <h1>List of Players</h1>
      <hr />
      <ListofPlayers players={players} />
      <hr />
      <h1>List of Players having Scores Less than 70</h1>
      <hr />
      <ListofPlayers players={playersBelow70} />
    </div>
  );
} else {
  return (
    <div className="App">
      <h1>Indian Team</h1>
      <hr />
      <h1>Odd Players</h1>
      <OddPlayers IndianTeam={IndianTeam} />
      <hr />
      <h1>Even Players</h1>
      <EvenPlayers IndianTeam={IndianTeam} />
      <hr />
      <h1>List of Indian Players Merged:</h1>
      <hr />
      <ListofIndianPlayers IndianPlayers={IndianTeam} />
    </div>
  );
}
}

export default App;

```

Output:

When Flag=true

List of Players	
•	Mr. Jack 50
•	Mr. Michael 70
•	Mr. John 40
•	Mr. Ann 61
•	Mr. Elisabeth 61
•	Mr. Sachin 95
•	Mr. Dhoni 100
•	Mr. Virat 84
•	Mr. Jadeja 64
•	Mr. Raina 75
•	Mr. Rohit 80

List of Players having Scores Less than 70	
•	Mr. Jack 50
•	Mr. Michael 70
•	Mr. John 40
•	Mr. Ann 61
•	Mr. Elisabeth 61
•	Mr. Jadeja 64

When Flag=false

Indian Team	
Odd Players	
•	First: First Player
•	Third: Third Player
•	Fifth: Fifth Player

Even Players	
•	Second: Second Player
•	Fourth: Fourth Player
•	Sixth: Sixth Player

List of Indian Players Merged:	
•	Mr. First Player
•	Mr. Second Player
•	Mr. Third Player
•	Mr. Fourth Player
•	Mr. Fifth Player
•	Mr. Sixth Player

10. ReactJS-HOL

Objectives

- Define JSX
- Explain about ECMA Script
- Explain React.createElement()
- Explain how to create React nodes with JSX
- Define how to render JSX to DOM
- Explain how to use JavaScript expressions in JSX
- Explain how to use inline CSS in JSX

In this hands-on lab, you will learn how to:

- Use JSX syntax in React applications
- Use inline CSS in JSX

Prerequisites

The following is required to complete this hands-on lab:

- js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **60 minutes.**

Create a React Application named "officespacerentalapp" which uses React JSX to create elements, attributes and renders DOM to display the page.

Create an element to display the heading of the page.

Attribute to display the image of the office space

Create an object of office to display the details like Name, Rent and Address.

Create a list of Object and loop through the office space item to display more data.

To apply Css, Display the color of the Rent in Red if it's below 60000 and in Green if it's above 60000.

App.js

```

import React from 'react';
import './App.css';

function App() {
  const element = "Office Space";

  const jsxatt = ;

  const ItemName = {
    Name: "DBS",
    Rent: 50000,
    Address: "Chennai"
  };

  const offices = [
    { Name: "Tech Hub", Rent: 75000, Address: "Bangalore" },
    { Name: "Innovate Co.", Rent: 45000, Address: "Hyderabad" },
    { Name: "City Works", Rent: 60000, Address: "Mumbai" },
    { Name: "Global HQ", Rent: 90000, Address: "Delhi" },
  ];

  return (
    <div className="App">
      <h1>{element}, at Affordable Range</h1>
      {jsxatt}

      <h1>Name: {ItemName.Name}</h1>
      <h3 className={ItemName.Rent <= 60000 ? 'textRed' : 'textGreen'}>
        Rent: Rs. {ItemName.Rent}
      </h3>
      <h3>Address: {ItemName.Address}</h3>

      <hr />
    </div>
  );
}

```



```

    {offices.map((office, index) => (
      <div key={index}>
        <h1>Name: {office.Name}</h1>
        <h3 className={office.Rent <= 60000 ? 'textRed' : 'textGreen'}>
          Rent: Rs. {office.Rent}
        </h3>
        <h3>Address: {office.Address}</h3>
        {index < offices.length - 1 && <hr />}
      </div>
    ))}
  </div>
);
}

export default App;

```

App.css

```

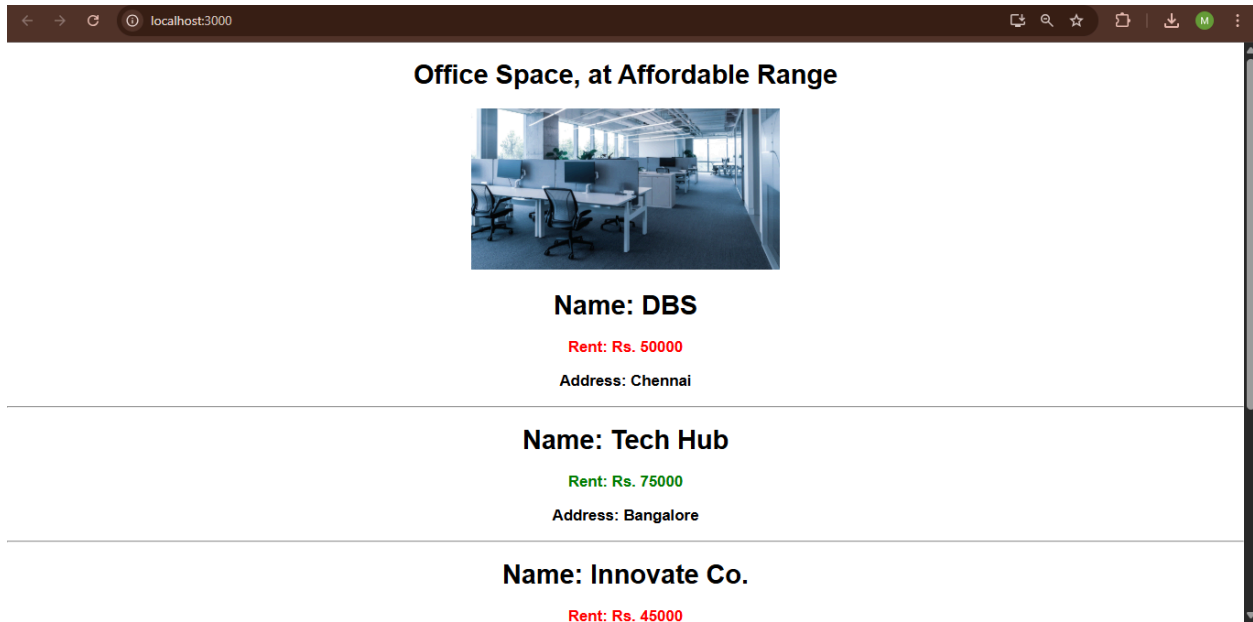
.App {
  font-family: sans-serif;
  text-align: center;
}

.textRed {
  color: red;
}

.textGreen {
  color: green;
}

```

Output:



11. ReactJS-HOL

Objectives

- Explain React events
- Explain about event handlers
- Define Synthetic event
- Identify React event naming convention

In this hands-on lab, you will learn how to:

- Implement Event handling concept in React applications
- Use this keyword
- Use synthetic event

Prerequisites

The following is required to complete this hands-on lab:

- js

- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **90 minutes**.

Create a React Application "eventexamplesapp" to handle various events of the form elements in HTML.

App.js

```
import React, { useState } from 'react';
import './App.css';

function App() {
  const [count, setCount] = useState(0);

  const [amount, setAmount] = useState('');
  const [currency, setCurrency] = useState('');

  const handleIncrement = () => {
    setCount(prevCount => prevCount + 1);
  };

  const sayHello = () => {
    alert('Hello! Member1');
  };

  const handleMultipleActions = () => {
    handleIncrement();
    sayHello();
  };

  const handleDecrement = () => {
    setCount(prevCount => prevCount - 1);
  };
}
```

```

const sayWelcome = (message) => {
  alert(message);
};

const handleClickMe = (event) => {
  alert('I was clicked');
  console.log('Synthetic Event:', event);
};

const handleAmountChange = (e) => {
  setAmount(e.target.value);
};

const handleCurrencyChange = (e) => {
  setCurrency(e.target.value);
};

const handleSubmit = (e) => {
  e.preventDefault();

  const exchangeRate = 80;
  let convertedAmount = 0;

  if (currency.toLowerCase() === 'rupees') {
    convertedAmount = parseFloat(amount) / exchangeRate;
    alert(`${amount} Indian Rupees (INR) is ${convertedAmount.toFixed(2)} Euro`);
  } else if (currency.toLowerCase() === 'euro') {

    convertedAmount = parseFloat(amount) * exchangeRate;
    alert(`${amount} Euro is ${convertedAmount.toFixed(2)} Indian Rupees (INR)`);

  } else {
    alert('Please enter "Euro" or "Rupees" as the target currency for conversion');
  }
};

```

```

n.');
```

```

  }
};

return (
  <div className="App">
    <h1>Counter: {count}</h1>
    <button onClick={handleMultipleActions}>Increment</button>
    <button onClick={handleDecrement}>Decrement</button>
    <br /><br />

    <button onClick={() => sayWelcome('Welcome')}>Say Welcome</button>
    <br /><br />

    <button onClick={handleClickMe}>Click on me</button>
    <br /><br />

    <h2>Currency Convertor!!!</h2>
    <form onSubmit={handleSubmit}>
      <div>
        <label htmlFor="amount">Amount:</label>
        <input
          type="number"
          id="amount"
          value={amount}
          onChange={handleAmountChange}
          required
        />
      </div>
      <div>
        <label htmlFor="currency">Currency:</label>
        <input
          type="text"
          id="currency"
          value={currency}
          onChange={handleCurrencyChange}

```

```

        placeholder="e.g., Euro or Rupees"
        required
      />
    </div>
    <button type="submit">Convert</button>
  </form>
</div>
);
}

export default App;

```

App.css

```

.App {
  font-family: sans-serif;
  text-align: left;
  margin: 20px;
}

button {
  margin: 5px;
  padding: 10px 20px;
  font-size: 16px;
  cursor: pointer;
}

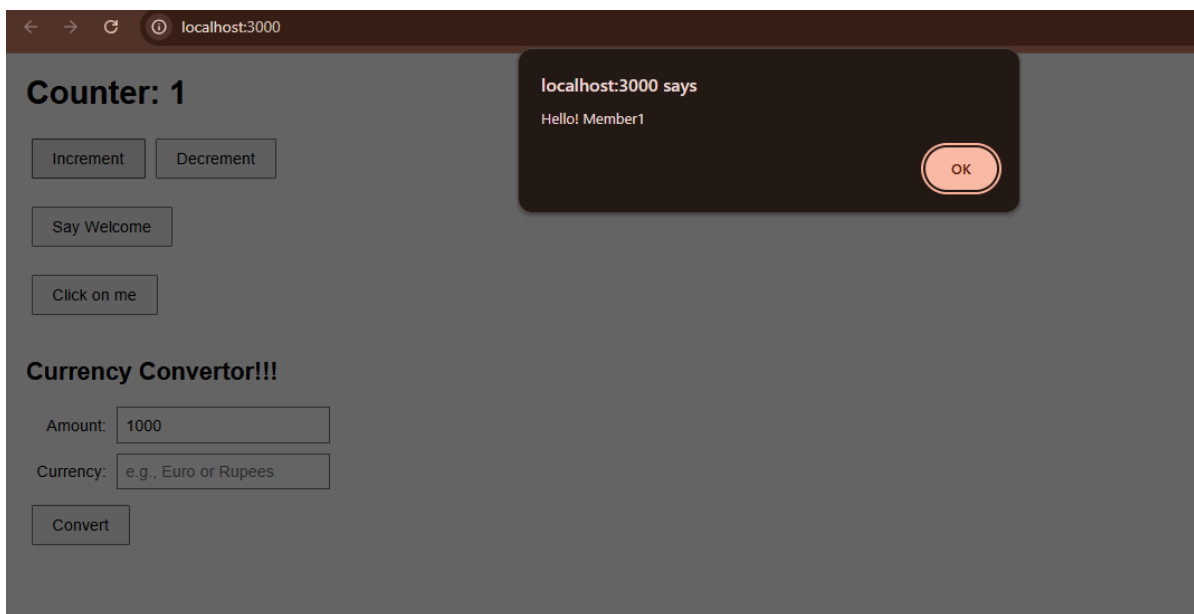
form div {
  margin-bottom: 10px;
}

form label {
  display: inline-block;
  width: 80px;
  text-align: right;
  margin-right: 10px;
}

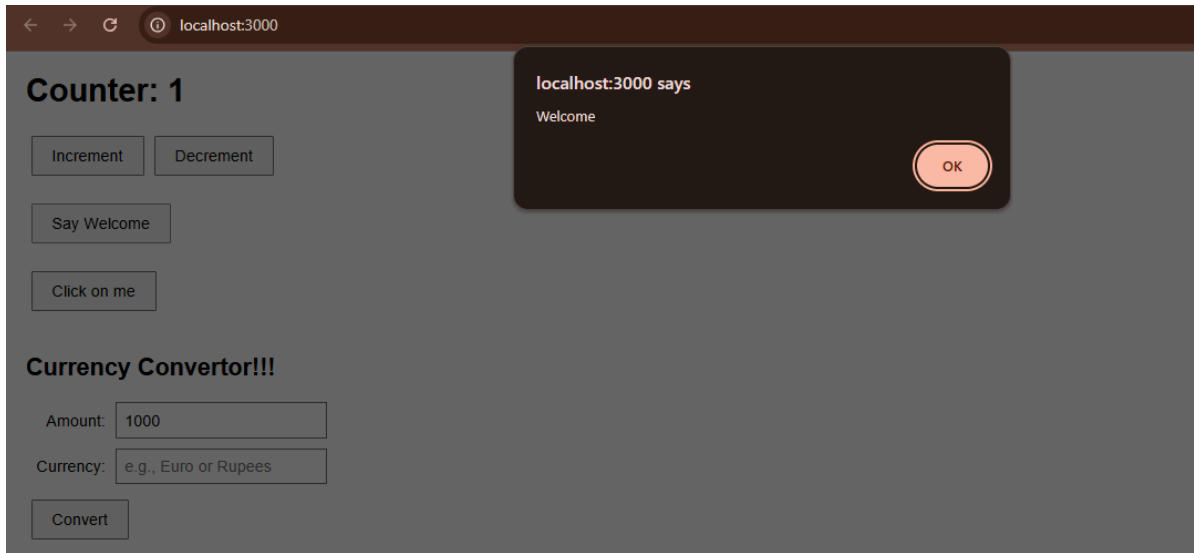
```

```
}  
  
form input {  
  padding: 8px;  
  font-size: 16px;  
}
```

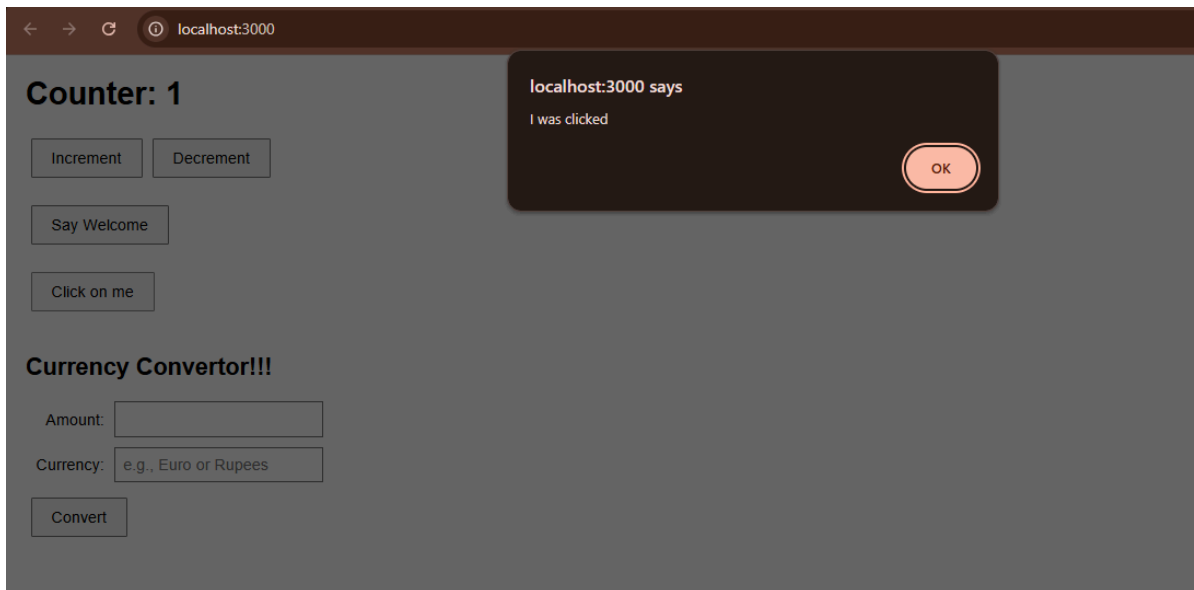
1. Create "Increment" button to increase the value of the counter and "Decrement" button to decrease the value of the counter. The "Increase" button should invoke multiple methods.
 - a. To increment the value
 - b. Say Hello followed by a static message.



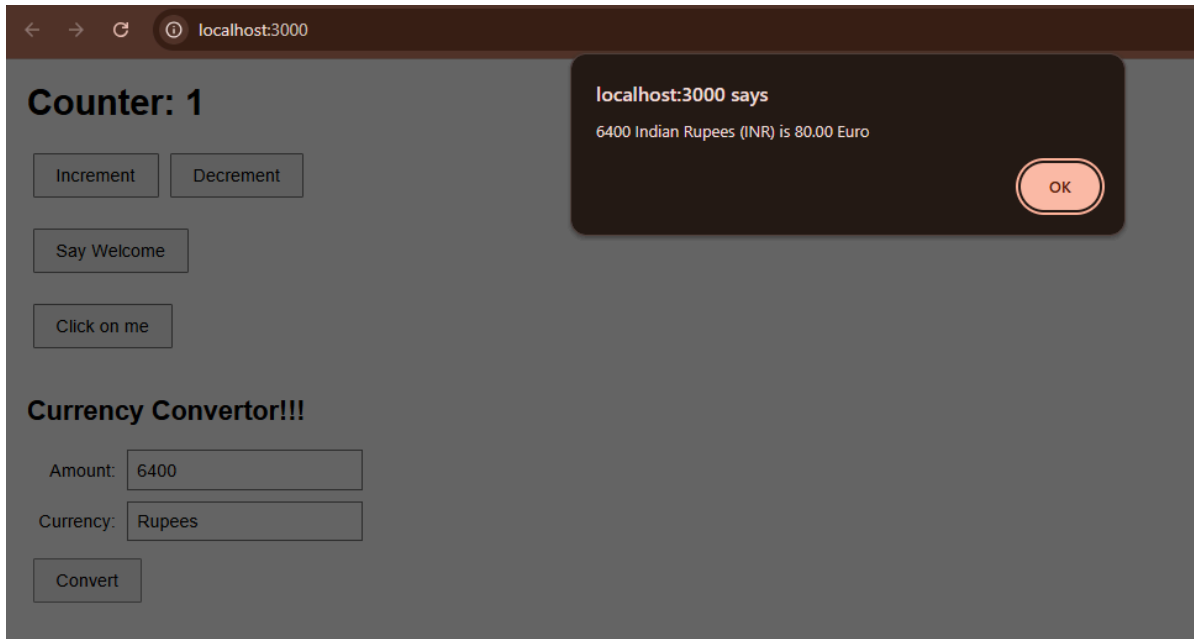
2. Create a button "Say Welcome" which invokes the function which takes "welcome" as an argument.



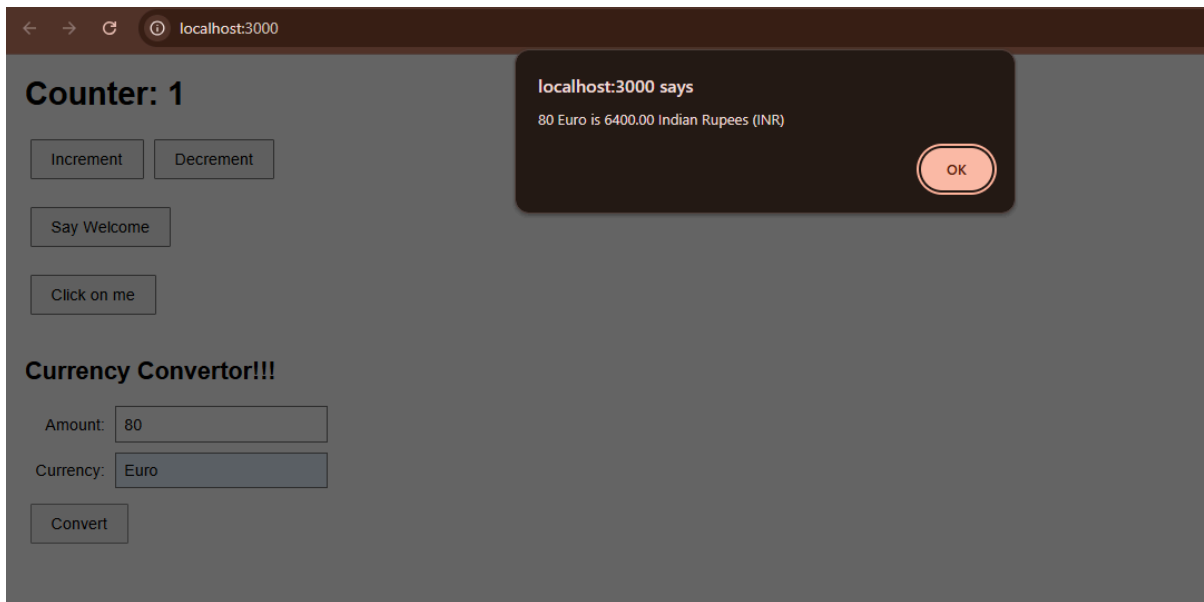
3. Create a button which invokes synthetic event "OnPress" which display "I was clicked"



4. Create a "CurrencyConvertor" component which will convert the Indian Rupees to Euro when the Convert button is clicked.



Handle the Click event of the button to invoke the handleSubmit event and handle the conversion of the euro to rupees.



12. ReactJS-HOL

Objectives

- Explain about conditional rendering in React
- Define element variables
- Explain how to prevent components from rendering

In this hands-on lab, you will learn how to:

- Implement conditional rendering in React applications

Prerequisites

The following is required to complete this hands-on lab:

- js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **60 minutes**.

Create a React Application named "ticketbookingapp" where the guest user can browse the page where the flight details are displayed whereas the logged in user only can book tickets.

The Login and Logout buttons should accordingly display different pages. Once the user is logged in the User page should be displayed. When the user clicks on Logout, the Guest page should be displayed.

App.js

```
import React, { useState } from 'react';
import './App.css';

// Component to display a greeting for a logged-in user
function UserGreeting() {
  return <h1>Welcome back</h1>;
}

// Component to display a greeting for a guest user
```

```

function GuestGreeting() {
  return <h1>Please sign up.</h1>;
}

// Component that handles conditional rendering of greetings
function Greeting(props) {
  const isLoggedIn = props.isLoggedIn;
  if (isLoggedIn) {
    return <UserGreeting />;
  }
  return <GuestGreeting />;
}

// Button component for logging in
function LoginButton(props) {
  return (
    <button onClick={props.onClick}>
      Login
    </button>
  );
}

// Button component for logging out
function LogoutButton(props) {
  return (
    <button onClick={props.onClick}>
      Logout
    </button>
  );
}

// Main App component
function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false); // Initial state is logged
  out

```

```
const handleLoginClick = () => {
  setIsLoggedIn(true);
};

const handleLogoutClick = () => {
  setIsLoggedIn(false);
};

let button;
if (isLoggedIn) {
  button = <LogoutButton onClick={handleLogoutClick} />;
} else {
  button = <LoginButton onClick={handleLoginClick} />;
}

return (
  <div className="App">
    <Greeting isLoggedIn={isLoggedIn} />
    {button}
  </div>
);
}

export default App;
```



Please sign up.

Login



Welcome back

Logout

13. ReactJS-HOL

Objectives

- Explain various ways of conditional rendering
- Explain how to render multiple components
- Define list component
- Explain about keys in React applications

- Explain how to extract components with keys
- Explain React Map, map() function

In this hands-on lab, you will learn how to:

- Implement conditional rendering in React applications

Prerequisites

The following is required to complete this hands-on lab:

- js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **60 minutes**.

Create a React App named "bloggerapp" in with 3 components.

1. Book Details
2. Blog Details
3. Course Details

Implement this with as many ways possible of Conditional Rendering.

App.js

```
import React, { useState } from 'react';
import './App.css';

// --- Data Definitions ---
const books = [
  { id: 101, bname: 'Master React', price: 670 },
  { id: 102, bname: 'Deep Dive into Angular 11', price: 800 },
  { id: 103, bname: 'Mongo Essentials', price: 450 },
];

const blogs = [
```

```

    { id: 201, title: 'React Learning', author: 'Stephen Biz', content: 'Welcome to learning React!' },
    { id: 202, title: 'Installation', author: 'Schewzdenier', content: 'You can install React from npm.' },
  ];

const courses = [
  { id: 301, name: 'Angular', date: '4/5/2021' },
  { id: 302, name: 'React', date: '6/3/2020' },
];

// --- Component Definitions ---

// Book Details Component
function BookDetails({ books, showBooks }) {
  // Conditional Rendering Method 1: If-else statement
  if (!showBooks) {
    return null; // Don't render anything if showBooks is false
  }
  return (
    <div className="st2">
      <h1>Book Details</h1>
      <ul>
        {books.map(book => (
          <li key={book.id}>
            <h3>{book.bname}</h3>
            <h4>{book.price}</h4>
          </li>
        ))}
      </ul>
    </div>
  );
}

// Blog Details Component
function BlogDetails({ blogs, showBlogs }) {

```

```
// Conditional Rendering Method 2: Logical && operator
return (
  showBlogs && (
    <div className="v1">
      <h1>Blog Details</h1>
      {blogs.length > 0 ? ( // Conditional Rendering Method 3: Ternary operator
        <ul>
          {blogs.map(blog => (
            <li key={blog.id}>
              <h3>{blog.title}</h3>
              <h4>{blog.author}</h4>
              <p>{blog.content}</p>
            </li>
          ))}
        </ul>
      ) : (
        <p>No blog posts available.</p>
      )}
    </div>
  )
);
}
```

```
// Course Details Component
function CourseDetails({ courses, showCourses }) {
  // Conditional Rendering Method 4: Element Variables
  let courseContent;
  if (showCourses) {
    courseContent = (
      <div className="mystyle1">
        <h1>Course Details</h1>
        <ul>
          {courses.map(course => (
            <li key={course.id}>
              <h3>{course.name}</h3>
              <h4>{course.date}</h4>
            </li>
          ))}
        </ul>
      </div>
    );
  }
}
```



```

    </li>
  )))
</ul>
</div>
);
} else {
  courseContent = null; // Or a placeholder if you prefer
}
return courseContent;
}

```

// Main App Component

```

function App() {
  // State variables to control conditional rendering
  const [showBooks, setShowBooks] = useState(true);
  const [showBlogs, setShowBlogs] = useState(true);
  const [showCourses, setShowCourses] = useState(true);

  return (
    <div className="App">
      <div className="controls">
        <button onClick={() => setShowBooks(!showBooks)}>
          Toggle Book Details
        </button>
        <button onClick={() => setShowBlogs(!showBlogs)}>
          Toggle Blog Details
        </button>
        <button onClick={() => setShowCourses(!showCourses)}>
          Toggle Course Details
        </button>
      </div>
      <div className="details-container">
        <BookDetails books={books} showBooks={showBooks} />
        <BlogDetails blogs={blogs} showBlogs={showBlogs} />
        <CourseDetails courses={courses} showCourses={showCourses} />
      </div>
    </div>
  );
}

```

```

    </div>
  );
}

export default App;

```

App.css

```

.App {
  font-family: sans-serif;
  text-align: center;
  padding: 20px;
}

.controls button {
  margin: 10px;
  padding: 10px 15px;
  font-size: 16px;
  cursor: pointer;
}

.details-container {
  display: flex;
  justify-content: space-around;
  margin-top: 30px;
}

.st2, .v1, .mystyle1 {
  border-left: 3px solid green; /* Vertical line as in the image */
  padding-left: 20px;
  margin: 0 10px;
  text-align: left;
  flex: 1; /* Distribute space evenly */
}

ul {

```

```
list-style-type: none;
padding: 0;
}

li {
margin-bottom: 15px;
}

h1 {
color: #333;
}

h3 {
margin-bottom: 5px;
}

h4 {
margin-top: 0;
color: #555;
}

p {
font-size: 0.9em;
color: #666;
}
```

Output:

Book Details

Master React
670

Deep Dive into Angular 11
800

Mongo Essentials
450

Blog Details

React Learning
Stephen Biz

Welcome to learning React!

Installation
Schewzdenier

You can install React from npm.

Course Details

Angular
4/5/2021

React
6/3/2020

Blog Details

React Learning
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6/3/2020